Bias, variance and cross-validation

Data science · University · Lecture 2



In this lecture

- Decompose expected error.
- Estimate error with cross-validation.
- Prevent leakage in student work.

Remember

- Error = bias^2 + variance + noise.
- Choose on unseen data.
- Preprocess inside each fold.

Find the leak

1. Warm-up (5 min): Which model wins?
2. Teach (15 min): The U-curve.
3. Hunt (20 min): Three notebooks.
4. Fix (15 min): A pipeline.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. Why does training error always fall?
2. What is the unit of independence in your data?
3. How will your brief stop leakage?

Exit ticket

- Name one leak and the line of code that fixes it.